

Lecture 05

Extreme Event Analysis-Part II (Flood Frequency Analysis & POT Approach)

Review on Lecture 4

- ❖ Extreme Event Analysis: Focusing on Floods!
- ❖ Questions to be answered in today's lecture:
 - ◆ What distribution to use?
 - ◆ What design event to use?
- ❖ Distributions for Extreme Values
 - ◆ Three Types of Extreme Distributions
 - Gumbel, Fréchet, Weibull (Type I, Type II, Type III)
 - ◆ Generalized Extreme Value (GEV) Distribution

Today's Topic: Lecture 05

❖ Flood Frequency Analysis

- ◆ Empirical Method (No Distributional Assumptions)
- ◆ Analytical Method (Distributional Assumptions)

❖ More Distributions for Extreme Events

- ◆ Generalized Extreme Value (GEV) Distributions
- ◆ Peaks-over-threshold (POT) Approach (=Partial Duration)
 - Generalized Pareto Distribution (GPD)
- ◆ How to Choose the Threshold u ?
 - Mean Residual Life Plot
 - Parameter Stability Plot

Flood Frequency Analysis (Haan Ch.7)

❖ Flood Frequency Analysis (홍수빈도분석)

◆ GOALS

- Estimate the magnitude of an event having a given frequency of occurrence (Return Period $T \rightarrow$ Design Event X_T)
- To estimate the frequency of an event having a given magnitude (Design Event $X_T \rightarrow$ Return Period T)

◆ Remember the following questions:

- For a specific type of structure, what should T be?
- Once we do choose T , how should we find X_T ?
 - Empirical Method (No Distributional Assumptions)
 - Analytical Method (Distributional Assumptions - GEV, LP3, GPD, etc.)

Flood Frequency Analysis (Haan Ch.7)

❖ Two Methods

- ◆ Empirical Method (No Distributional Assumptions)
- ◆ Analytical Method (Distributional Assumptions)

❖ Empirical Method includes

- ◆ Probability Plotting
- ◆ Probability papers for different distributions:
Normal, Lognormal, Exponential, Gamma, GEV, Weibull, etc.

Flood Frequency Analysis: (1) Empirical

❖ Procedures

- (1) Rank the data from the smallest (or largest) to the largest (or smallest) value.
- (2) Calculate the plotting positions of each values.
- (3) Select the type of probability paper to be used.
- (4) Plot the observations on the probability paper.

TABLE 18.3.1 Alternative Plotting Positions and their Motivation*

Name	Formula	a	T_1	Motivation
Weibull	$\frac{i}{n+1}$	0	$n+1$	Unbiased exceedance probabilities for all distributions
Median†	$\frac{i-0.3175}{n+0.365}$	0.3175	$1.47n+0.5$	Median exceedance probabilities for all distributions
APL	$\frac{i-0.35}{n}$	~ 0.35	$1.54n$	Used with PWMs [Eq. (18.1.13)]
Blom	$\frac{i-3/8}{n+1/4}$	0.375	$1.60n+0.4$	Unbiased normal quantiles
Cunnane	$\frac{i-0.40}{n+0.2}$	0.40	$1.67n+0.3$	Approximately quantile-unbiased
Gringorten	$\frac{i-0.44}{n+0.12}$	0.44	$1.79n+0.2$	Optimized for Gumbel distribution
Hazen	$\frac{i-0.5}{n}$	0.50	$2n$	A traditional choice

* Here a is the plotting-position parameter in Eq. (18.3.6) and T_1 is the return period each plotting position assigns to the largest observation in a sample of size n .

† For $i=1$ and n , the exact value is $q_1 = 1 - q_n = 1 - 0.5^{1/n}$.

TABLE 18.3.2 Generation of Probability Plots for Different Distributions

Normal probability paper. Plot $x_{(i)}$ versus z_{p_i} given in Eq. (18.2.3), where $p_i = 1 - q_i$. Blom's formula ($a = 3/8$) provides quantile-unbiased plotting positions.

Lognormal probability paper. Plot ordered logarithms $\log [x_{(i)}]$ versus z_{p_i} . Blom's formula ($a = 3/8$) provides quantile-unbiased plotting positions.

Exponential probability paper. Plot ordered observations $x_{(i)}$ versus $\xi - \ln(q_i)/\beta$ or just $-\ln(q_i)$. Gringorten's plotting positions ($a = 0.44$) work well.

Gumbel and Weibull probability paper. For Gumbel distribution plot ordered observations $x_{(i)}$ versus $\xi - \alpha \ln [-\ln(1 - q_i)]$ or just $y_i = -\ln [-\ln(1 - q_i)]$. Gringorten's plotting positions ($a = 0.44$) were developed for this distribution. For Weibull distribution plot $\ln [x_{(i)}]$ versus $\ln(\alpha) + \ln [-\ln(q_i)]/k$ or just $\ln [-\ln(q_i)]$. (See Ref. 154.)

GEV distribution. Plot ordered observations $x_{(i)}$ versus $\xi + (\alpha/\kappa) \{1 - [-\ln(1 - q_i)]^\kappa\}$, or just $(1/\kappa) \{1 - [-\ln(1 - q_i)]^\kappa\}$. Alternatively employ Gumbel probability paper on which GEV will be curved. Cunnane's plotting positions ($a = 0.4$) are reasonable.⁵²

Pearson type 3 probability paper. Plot ordered observations $x_{(i)}$ versus $K_p(\gamma)$, where $p_i = 1 - q_i$. Blom's formula ($a = 3/8$) is quantile-unbiased for normal distribution and makes sense for small γ . Or employ normal probability paper. (See Ref. 158.)

Log Pearson type 3 probability paper. Plot ordered logarithms $\log [x_{(i)}]$ versus $K_p(\gamma)$ where $p_i = 1 - q_i$. Blom's formula ($a = 3/8$) makes sense for small γ . Or employ lognormal probability paper. (See Ref. 158.)

Uniform probability paper. Plot $x_{(i)}$ versus $1 - q_i$, where q_i are the Weibull plotting positions ($a = 0$). (See Ref. 154.)

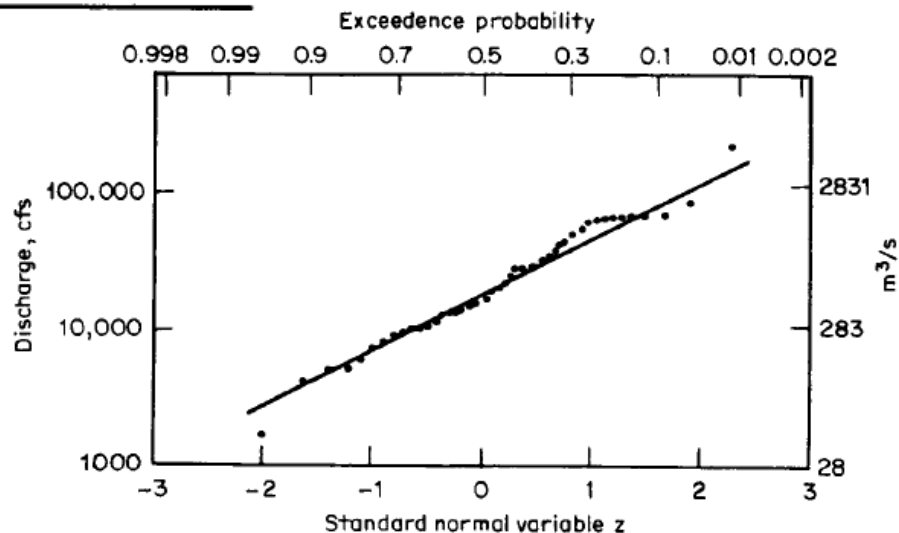


FIGURE 18.3.1 A probability plot using a normal scale of 44 annual maxima for the Guadalupe River near Victoria, Texas. (Reproduced with permission from Ref. 20, p. 398.)

Flood Frequency Analysis: (1) Empirical

❖ Example

Table 7.2. Determination of plotting position for Kentucky River data

Flow	Rank	pp	Flow	Rank	pp	Flow	Rank	pp	Flow	Rank	pp
144000	1	0.01	80400	26	0.26	65700	51	0.51	48800	76	0.76
115000	2	0.02	80100	27	0.27	63800	52	0.52	47300	77	0.77
112000	3	0.03	80000	28	0.28	62600	53	0.53	47000	78	0.78
111000	4	0.04	79600	29	0.29	62600	54	0.54	46800	79	0.79
107000	5	0.05	79200	30	0.3	61800	55	0.55	46300	80	0.8
105000	6	0.06	77900	31	0.31	61300	56	0.56	46100	81	0.81
101000	7	0.07	77700	32	0.32	60300	57	0.57	46000	82	0.82
99100	8	0.08	77000	33	0.33	60200	58	0.58	45000	83	0.83
98400	9	0.09	73400	34	0.34	58600	59	0.59	44200	84	0.84
96100	10	0.1	72900	35	0.35	58000	60	0.6	44000	85	0.85
94300	11	0.11	71700	36	0.36	58000	61	0.61	43100	86	0.86
93700	12	0.12	71200	37	0.37	55200	62	0.62	42900	87	0.87
92500	13	0.13	71000	38	0.38	55000	63	0.63	42100	88	0.88
91500	14	0.14	70800	39	0.39	54600	64	0.64	40600	89	0.89
89400	15	0.15	70500	40	0.4	54400	65	0.65	40300	90	0.9
88900	16	0.16	70000	41	0.41	53600	66	0.66	38900	91	0.91
87200	17	0.17	69900	42	0.42	53600	67	0.67	36500	92	0.92
87100	18	0.18	69000	43	0.43	53500	68	0.68	35100	93	0.93
85000	19	0.19	68800	44	0.44	53400	69	0.69	34700	94	0.94
84300	20	0.2	68700	45	0.45	52400	70	0.7	34100	95	0.95
84100	21	0.21	68400	46	0.46	52300	71	0.71	32300	96	0.96
82900	22	0.22	67800	47	0.47	52100	72	0.72	29400	97	0.97
82900	23	0.23	67200	48	0.48	51700	73	0.73	28400	98	0.98
82500	24	0.24	66900	49	0.49	50500	74	0.74	20600	99	0.99
80900	25	0.25	66300	50	0.5	49600	75	0.75			

Flood Frequency Analysis: (1) Empirical

❖ Example

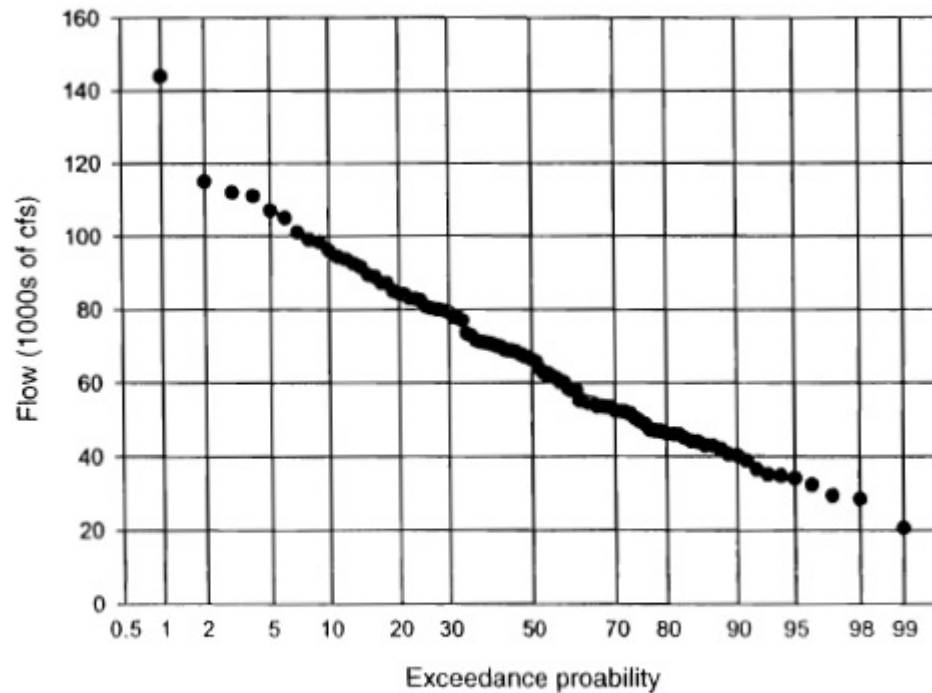
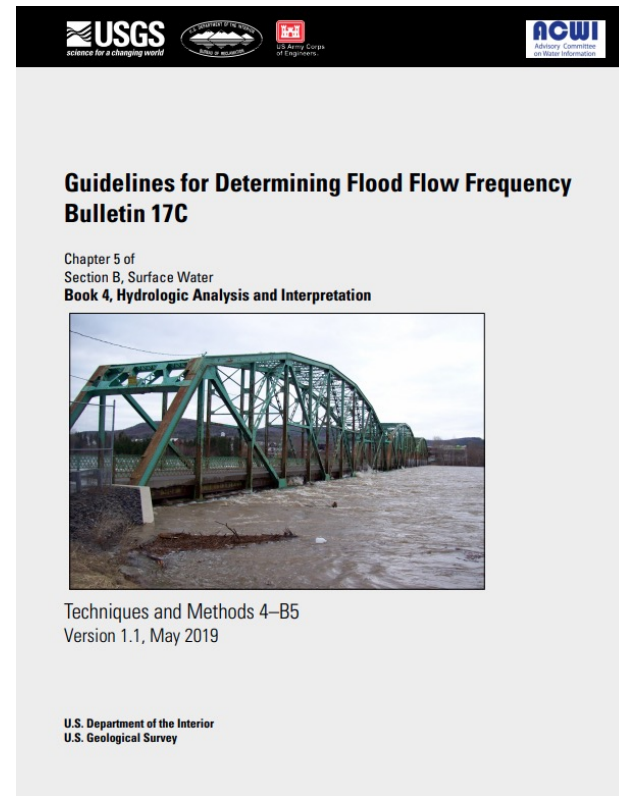


Fig. 7.1. Normal probability plot of Kentucky River flow data.

Flood Frequency Analysis: (2) Analytical

❖ 국내/미국의 설계홍수량 산정방법 (참고자료)

- ◆ 한국: 홍수량 산정 표준지침 (환경부)
- ◆ 미국: Bulletin 17B/17C (USGS: US Geological Survey)



국내 설계홍수량 산정방법



1.4 홍수량 산정방법

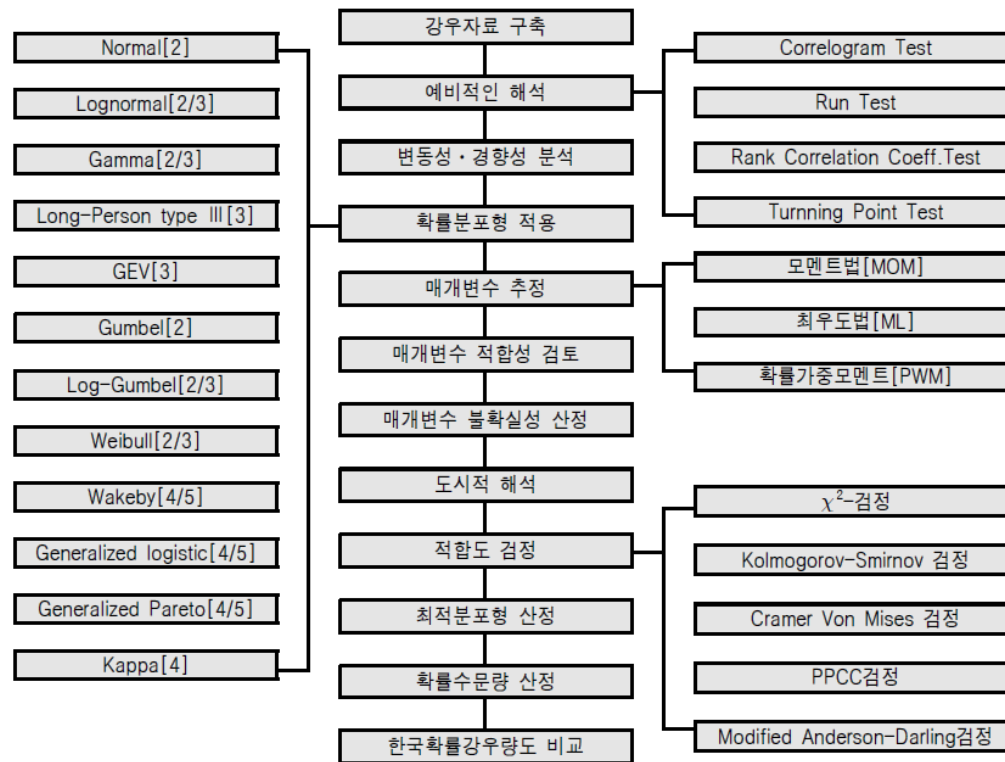
(1) 홍수량 산정방법은 크게 홍수량 빈도해석 방법과 강우-유출관계 모형에 의한 방법으로 구분한다.

해설

- (1) 홍수량 산정방법은 홍수량 빈도해석 방법과 강우-유출관계 모형에 의한 방법으로 나눌 수 있다.
- (2) 계속유역의 실측 연최대치계열의 홍수량 자료가 충분히 축적되어 있거나, 향후 축적될 경우 홍수량 산정 시 홍수량 직접빈도해석 결과와 강우-유출관계 모형, 기왕 홍수자료, 홍수흔적, 경험공식 등을 비교·검토하여 홍수량을 결정할 수 있다.
- (3) 홍수량 빈도해석에 의한 방법은 홍수량 자료가 충분히 계속된 유역에 대하여 연최대홍수량 자료 계열을 작성 후 홍수빈도분석에 의해 확률홍수량을 산정하는 방법이다. 홍수량이 미계측된 유역에 이 방법을 적용하기 위해서는 홍수량 계속유역에 대한 홍수빈도 분석결과와 계속유역의 지형특성 인자간 상관관계를 이용하는 지역화(regionalization)방법을 사용하여 확률홍수량을 결정하게 된다. 자세한 산정절차는 하천설계기준(2018)의 설계수문량 설계편을 참고한다.
- (4) 홍수량 빈도해석에 의한 방법은 홍수량 자료를 직접 빈도해석하여 확률홍수량을 산정하며 이론적으로 가장 정확한 방법이라 할 수 있다. 하지만, 현재 시점에서 우리나라의 실측 연최대치계열의 홍수량 자료가 절대적으로 부족하고, 기존 수위-유량관계곡선의 정확도가 낮아 환산하는 과정에서 홍수량 자료 계열 자체의 신뢰도가 낮은 문제점이 있다. 또한, 홍수조절능력을 가진 댐이 설치되어 있는 경우의 홍수량 산정 시에는 댐 건설 이전의 홍수량 자료를 댐 건설 이후의 홍수량 자료로 환산하여야 하지만 간단한 경험공식(Myers-Garris 공식) 이외에는 적절한 환산방법이 없으므로 홍수량 결정에 문제점이 있다.

2.2 확률강우량 산정 및 강우강도식 유도

- (1) 확률강우량 산정방법은 지역빈도해석 방법을 권장하고 최적 확률분포형은 GEV 분포형으로 한다.
- (2) 도서·해안 등 지역빈도해석 수행 및 적용이 용이하지 않은 지역의 경우에는 지점빈도해석을 통해 확률강우량을 산정할 수 있으며, 지점빈도해석의 최적 확률분포형은 Gumbel 분포형으로 한다.
- (3) 임의의 지속기간에 해당하는 확률강우량은 강우강도식을 작성하여 산정하며, 상관계수가 높은 General형이나 전대수 다항식형을 채택한다.



<그림 2.2> 지점빈도해석 절차

미국 설계홍수량 산정방법

❖ In US, log-Pearson III distribution is recommended for annual maximum floods

◆ Recommended in 1967

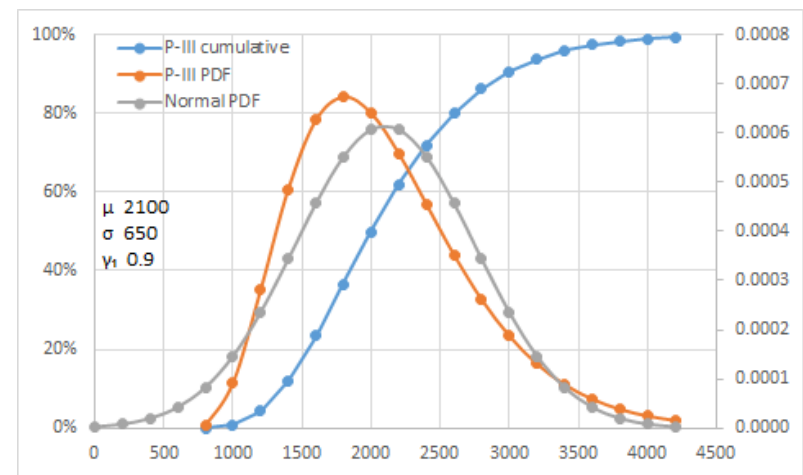
- “there are no rigorous statistical criteria on which to base a choice of method”
- “the range of uncertainty in flood analysis, regardless of the method used, is still large”

◆ Codified in 1976 (Bulletin 17)

◆ Updates: Bulletin 17B (1982) & Bulletin 17C (2018)

❖ Pearson Distributions

◆ A family of distributions (P1-P6)
to flexibly model
mean, variance, skew, kurtosis



미국 설계홍수량 산정방법

❖ Pearson Distributions

<Def>

Pearson type 3	$f_X(x) = \beta [\beta(x - \xi)]^{\alpha-1} \frac{\exp[-\beta(x - \xi)]}{\Gamma(\alpha)}$	$\alpha > 0$	$\mu_x = \xi + \frac{\alpha}{\beta}; \sigma_x^2 = \frac{\alpha}{\beta^2}$
	$\Gamma(\alpha)$ is the gamma function	for $\beta > 0: x > \xi$	and $\gamma_x = \frac{2}{\sqrt{\alpha}}$
	(for $\beta > 0$ and $\xi = 0: \gamma_x = 2 \text{ CV}_x$)	for $\beta < 0: x < \xi$	and $\gamma_x = \frac{-2}{\sqrt{\alpha}}$
log-Pearson type 3	$f_X(x) = \beta [\beta(\ln(x) - \xi)]^{\alpha-1} \frac{\exp(-\beta(\ln(x) - \xi))}{x\Gamma(\alpha)}$		See Eq. (18.2.34)
	for $\beta < 0, 0 < x < \exp(\xi)$; for $\beta > 0, \exp(\xi) < x < \infty$		

Model Estimation

✓ Method of Moments (MoM), recommended in 17B

$$\begin{aligned} X &= \log_{10}(Q) & \hat{M}_x &= \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \\ &\downarrow & \hat{\sigma}_x^2 &= s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \\ \hat{\alpha} &= 4/g_x^2 & \hat{\delta}_x &= g_x = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \frac{(x_i - \bar{x})^3}{s_x^3} \\ \hat{\beta} &= s_x g_x / 2 & & \\ \hat{\gamma} &= \bar{x} - 2 \cdot s_x / g_x & & \end{aligned}$$

↪ sampling variability ↑.

✓ Design Events

For p^{th} quantile (non-exceedance probability of p ,

ex. $X_{0.99} = 100\text{-yr flood}$) $\Rightarrow x_p = \bar{x} + s_x \boxed{K_p(g_x)}$

• K_p = frequency factor, find @ table.

the p^{th} quantile of P3 distribution with

$M=0$, $\sigma^2=1$, and skew g_x .

$$\Rightarrow \text{Flood flow } Q_p = 10^{x_p}$$

Flood Frequency Analysis: (2) Analytical

❖ Example #1: Normal Distribution

Flood Frequency Analysis: (2) Analytical

❖ Example #2: Lognormal Distribution

Flood Frequency Analysis: (2) Analytical

❖ Example #3: Log Pearson Type III Distribution

◆ Procedures

Table 7.3a. K_T values for positive skew coefficients Pearson type III distribution¹

Skew coef. γ	Recurrence interval (years)							
	1.0101	2	5	10	25	50	100	200
	Percent chance ($\geq$)							
	99	50	20	10	4	2	1	0.5
3.0	-0.667	-2.396	0.420	1.180	2.278	3.152	4.051	4.970
2.9	-0.690	-0.390	0.440	1.195	2.277	3.134	4.013	4.904
2.8	-0.714	-0.384	0.460	1.210	2.275	3.114	3.973	4.847
2.7	-0.740	-0.376	0.479	1.224	2.272	3.093	3.932	4.783
2.6	-0.769	-0.368	0.499	1.238	2.267	3.071	3.889	4.718
2.5	-0.799	-0.360	0.518	1.250	2.262	3.048	3.845	4.652
2.4	-0.832	-0.351	0.537	1.262	2.256	3.023	3.800	4.584
2.3	-0.867	-0.341	0.555	1.274	2.248	2.997	3.753	4.515
2.2	-0.905	-0.330	0.574	1.284	2.240	2.970	3.705	4.444
2.1	-0.946	-0.319	0.592	1.294	2.230	2.942	3.656	4.372
2.0	-0.990	-0.307	0.609	1.302	2.219	2.912	3.605	4.298
1.9	-1.037	-0.294	0.627	1.310	2.207	2.881	3.553	4.223
1.8	-1.087	-0.282	0.643	1.318	2.193	2.848	3.499	4.147
1.7	-1.140	-0.268	0.660	1.324	2.179	2.815	3.444	4.069
1.6	-1.197	-0.254	0.675	1.329	2.163	2.780	3.388	3.990
1.5	-1.256	-0.240	0.690	1.333	2.146	2.743	3.330	3.910
1.4	-1.318	-0.225	0.705	1.337	2.128	2.706	3.271	3.828
1.3	-1.383	-0.210	0.719	1.339	2.108	2.666	3.211	3.745
1.2	-1.449	-0.195	0.732	1.340	2.087	2.626	3.149	3.661
1.1	-1.518	-0.180	0.745	1.341	2.066	2.585	3.087	3.575
1.0	-1.588	-0.164	0.758	1.340	2.043	2.542	3.022	3.489
.9	-1.660	-0.148	0.769	1.339	2.018	2.498	2.957	3.401
.8	-1.733	-0.132	0.780	1.336	1.993	2.453	2.891	3.312
.7	-1.806	-0.116	0.790	1.333	1.967	2.407	2.824	3.223
.6	-1.880	-0.099	0.800	1.328	1.939	2.359	2.755	3.132
.5	-1.955	-0.083	0.808	1.323	1.910	2.311	2.686	3.041
.4	-2.029	-0.066	0.816	1.317	1.880	2.261	2.615	2.949
.3	-2.104	-0.050	0.824	1.309	1.849	2.211	2.544	2.856
.2	-2.178	-0.033	0.830	1.301	1.818	2.159	2.472	2.763
.1	-2.252	-0.017	0.836	1.292	1.785	2.107	2.400	2.670
0	-2.326	0	0.842	1.282	1.751	2.054	2.326	2.576

¹ Interagency Advisory Committee on Water Data (1982).

Table 7.3b. K_T values for negative skew coefficients Pearson type III distribution¹

Skew coef γ	Recurrence interval (years)							
	1.0101	2	5	10	25	50	100	200
	Percent chance ($\geq$)							
	99	50	20	10	4	2	1	0.5
.0	-2.326	0	0.842	1.282	1.751	2.054	2.326	2.576
-.1	-2.400	0.017	0.846	1.270	1.716	2.000	2.252	2.482
-.2	-2.472	0.033	0.850	1.258	1.680	1.945	2.178	2.388
-.3	-2.544	0.050	0.853	1.245	1.643	1.890	2.104	2.294
-.4	-2.615	0.066	0.855	1.231	1.606	1.834	2.029	2.201
-.5	-2.686	0.083	0.856	1.216	1.567	1.777	1.955	2.108
-.6	-2.755	0.099	0.857	1.200	1.528	1.720	1.880	2.016
-.7	-2.824	0.116	0.857	1.183	1.488	1.663	1.806	1.926
-.8	-2.891	0.132	0.856	1.166	1.448	1.606	1.733	1.837
-.9	-2.957	0.148	0.854	1.147	1.407	1.549	1.660	1.749
-1.0	-3.022	0.164	0.852	1.128	1.366	1.492	1.588	1.664
-1.1	-3.087	0.180	0.848	1.107	1.324	1.435	1.518	1.581
-1.2	-3.149	0.195	0.844	1.086	1.282	1.379	1.449	1.501
-1.3	-3.211	0.210	0.838	1.064	1.240	1.324	1.383	1.424
-1.4	-3.271	0.225	0.832	1.041	1.198	1.270	1.318	1.351
-1.5	-3.330	0.240	0.825	1.018	1.157	1.217	1.256	1.282
-1.6	-3.388	0.254	0.817	0.994	1.116	1.166	1.197	1.216
-1.7	-3.444	0.268	0.808	0.970	1.075	1.116	1.140	1.155
-1.8	-3.499	0.282	0.799	0.945	1.035	1.069	1.087	1.097
-1.9	-3.553	0.294	0.788	0.920	0.996	1.023	1.037	1.044
-2.0	-3.605	0.307	0.777	0.895	0.959	0.980	0.990	0.995
-2.1	-3.656	0.319	0.765	0.869	0.923	0.939	0.946	0.949
-2.2	-3.705	0.330	0.752	0.844	0.888	0.900	0.905	0.907
-2.3	-3.753	0.341	0.739	0.819	0.855	0.864	0.867	0.869
-2.4	-3.800	0.351	0.725	0.795	0.823	0.830	0.832	0.833
-2.5	-3.845	0.360	0.711	0.771	0.793	0.798	0.799	0.800
-2.6	-3.889	0.368	0.696	0.747	0.764	0.768	0.769	0.769
-2.7	-3.932	0.376	0.681	0.724	0.738	0.740	0.740	0.741
-2.8	-3.973	0.384	0.666	0.702	0.712	0.714	0.714	0.714
-2.9	4.013	0.390	0.651	0.681	0.683	0.689	0.690	0.690
-3.0	4.051	0.396	0.636	0.660	0.666	0.666	0.667	0.667

¹ Interagency Advisory Committee on Water Data (1982).

General Discussions (Haan p.178-179)

1. Is the type of storm that is likely to produce the 100-year flow represented in the observed sample?
2. Is the contributing area of the basin the same for extreme floods as it is for small ones?
3. Are there ponds and reservoirs that may discharge at high rates during rare floods and not during smaller flows? What is the possibility of a dam breach and what would be the resulting flow?
4. Are the channel flow and storage characteristics the same for extreme flows as they are for smaller flows?
5. Are land use and soil characteristics such that flows from rare storms may relate to precipitation in a manner different from more common storms?
6. Are there seasonal effects such that rare floods are more likely to occur in a different season than the more common floods?
7. Is the rare flood represented in the sample of data? If so how is it treated? Is it assigned a return period of 15 years where in fact its return period may be much greater than that?
8. Are changes going on within the basin that may cause change in the hydrologic response of the basin to rainstorms?

9. Are there climatic changes occurring that may influence flood flow frequencies?

Peaks-Over-Threshold (POT) Models

❖ Peaks-over-Thresholds (POT) approach (=partial duration)

◆ Consider ALL extremes such that $x_i > u$, for some high threshold u

✓ Peaks-over-thresholds (POT) approach (=partial duration)

: consider all extremes $X_i > u$, for some high threshold u .

Assume $X_1, X_2, \dots$ are independent and identically distributed (iid) RVs with unknown CDF F_X . Then,

$$P(X > u+y | X > u) = \frac{P(X > u+y \cap X > u)}{P(X > u)} = \frac{1 - F_X(u+y)}{1 - F_X(u)}$$

- X is some frequently measured process (i.e. daily precip)
- We limit our analysis to large values of X (i.e. $X > u$)
- We want is the distribution of the exceedances beyond u , i.e., $P(X > u+y | X > u)$, $y = \text{exceedance}$.

$$P(Y > y | X > u) \text{ with } Y = X - u$$

★ Find limiting dist. for RV Y as n gets large.

* Generalized Pareto Distribution (GPD)

Recall : For $Z = \max(X_1, X_2, \dots, X_n)$ and n large,

$$Z \sim \text{GEV}(\mu, \sigma, \xi)$$

Then, for large enough values of u , exceedances

$Y = X - u$ given $X > u$ follow $Y \sim \text{GPD}(\xi, \tilde{\sigma})$

$$F_Y(y) = 1 - \left(1 + \frac{\xi y}{\tilde{\sigma}}\right)^{-1/\xi} \quad \text{w/} \quad \tilde{\sigma} = \sigma + \xi(u - \mu)$$

Note

- Parameters of GPD linked to those of GEV
- If $\xi < 0$, GPD has upper bound of $u - \tilde{\sigma}/\xi$
- If $\xi > 0$, no upper bound.
- If $\xi = 0$, $Y \sim \text{Exponential dist.}$

$$F_Y(y) = 1 - \exp(-y/\tilde{\sigma})$$

Selecting Threshold u

due Mon (10/21).

Selecting Threshold u

For large u , $X > u$, $Y = X - u$

$$F_Y(y) = \begin{cases} 1 - \left[1 + \frac{\xi y}{\hat{\sigma}}\right]^{-1/\xi} & (\xi \neq 0) \\ 1 - \exp(y/\hat{\sigma}) & (\xi = 0). \end{cases}$$

Bias-Variance Tradeoff

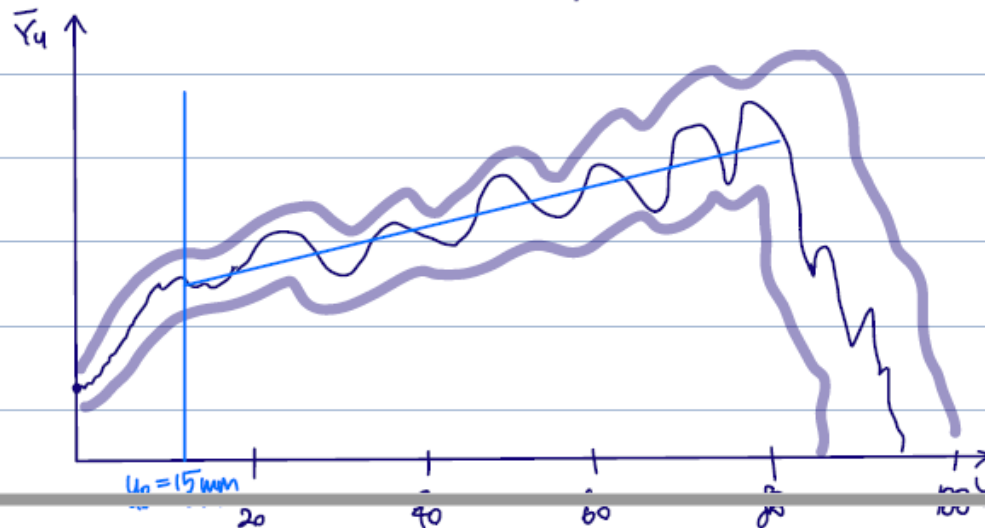
- u too small, GPD no longer limiting dist. of y (biased)
- u too big, very few exceedances, high variance of $\hat{\sigma}, \hat{\xi}$

* GOAL: choose my threshold big enough so $Y \sim \text{GPD}$,
but no bigger to maximize available data.

Selecting Threshold u

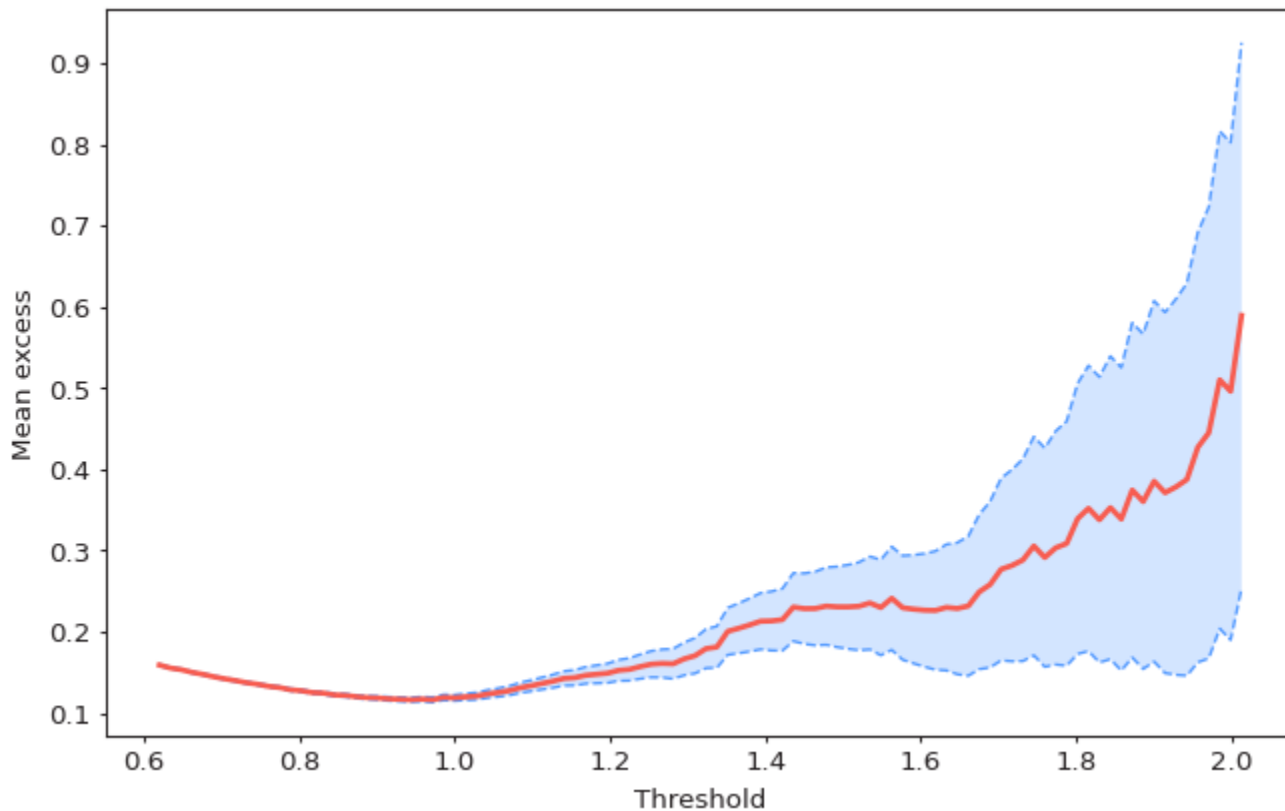
Procedure

- ① For values of u from $X_{\min}$ to just under $X_{\max}$, estimate the expected exceedance, $E[Y_u] = \bar{Y}_u = \frac{1}{n_u} \sum_{i=1}^{n_u} y_i$
- ② Plot $\bar{Y}_u$ vs. u
- ③ Add 95% C.I. for $\bar{Y}_u$ based on $\text{var}(\bar{Y}_u) = \frac{s_{y_n}^2}{n_u}$
- ④ Look for smallest u where $\bar{Y}_u$ vs. u is **linear**



Selecting Threshold u

❖ Example of Mean Residual Life Plot



Parameter Estimation & Declustering

- ❖ Given $\mathbf{u}$, estimate via MLE
- ❖ Analytical maximization of Log-Likelihood is hard,
thus use numerical optimization
- ❖ Declustering
 - ◆ GPD assumes independent exceedances
 - ◆ However, not usually true in hydrology
 - 3-day intense rainfall
 - Long lasting floods

Parameter Estimation & Declustering

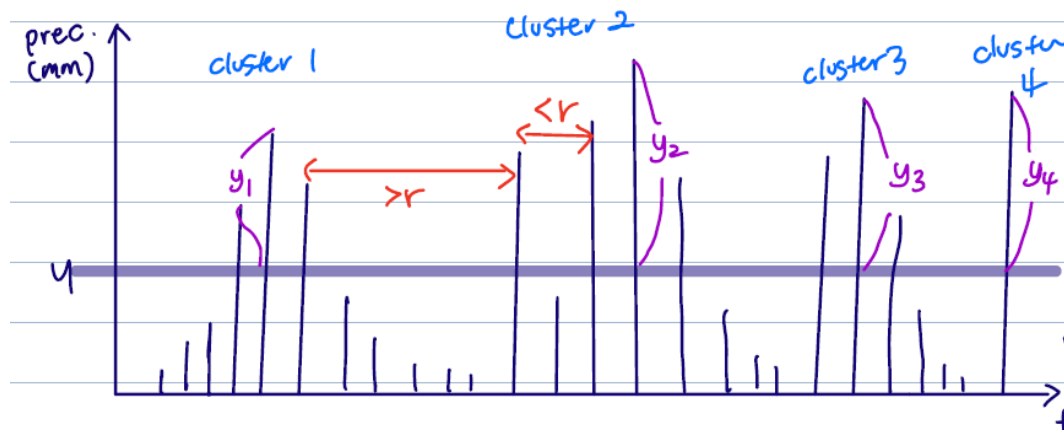
❖ Procedures for Declustering

- (1) For fixed u , find clusters of consecutive exceedances
- (2) Set run length r to separate clusters
- (3) Take maximum exceedance in each clusters

❖ Example:

Procedure

- ① For fixed u , find clusters of consecutive exceedances.
- ② Set run length r to separate clusters. $\Rightarrow r=3$
(depends on region)
- ③ take max. exceedance in each clusters.



Future Schedule Preview

❖ 다음 주 강의 (2022-11-01 화요일 18:30-21:20)

◆ *Training with Extreme Event Analysis*

- Flood Frequency Analysis using Extreme Distributions
- Using Peaks-Over-Threshold (POT) Models

❖ Other Assignments

◆ Datacamp Assignments (Assigned through Web)

- Introduction to Tidyverse due 10/28
- Cleaning Data in R due 11/04

◆ Mid-Term Project Report & Presentation: 11/08 Tue (Dinner?)

◆ TEST Schedule? 11/29 Tue (예정)

Tentative Schedule (~End of Semester)

WEEK	DATE	TOPIC
8	Oct-25 (Today)	Lec 05: Extreme Event Analysis Part II
9	Nov-01	Training with Extreme Event Analysis (Laptop)
10	Nov-08	Mid Term Project Presentations (& Dinner)
11	Nov-15	Quiz 2 / Lec 06: Introduction to Regression
12	Nov-22	Test Preparation (Example Problem Solutions)
13	Nov-29	TEST (Closed Book)
14	Dec-06	Lec 07: Hydrologic Statistics Frontiers
15	Dec-13	Final Project Presentation (Teams of 4, TBA)